PixelLogic

Complete Application Tutorial

Version 1.0.138

Professional LED wall design, pixel mapping, and AV engineering tool.

This tutorial covers all tabs and features with screenshots and detailed explanations.

Designed for AV technicians, LED engineers, and production designers.

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Generated: October 28, 2026

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16 sections • 36 total pages

1. Projects

Create & Manage Your LED Shows

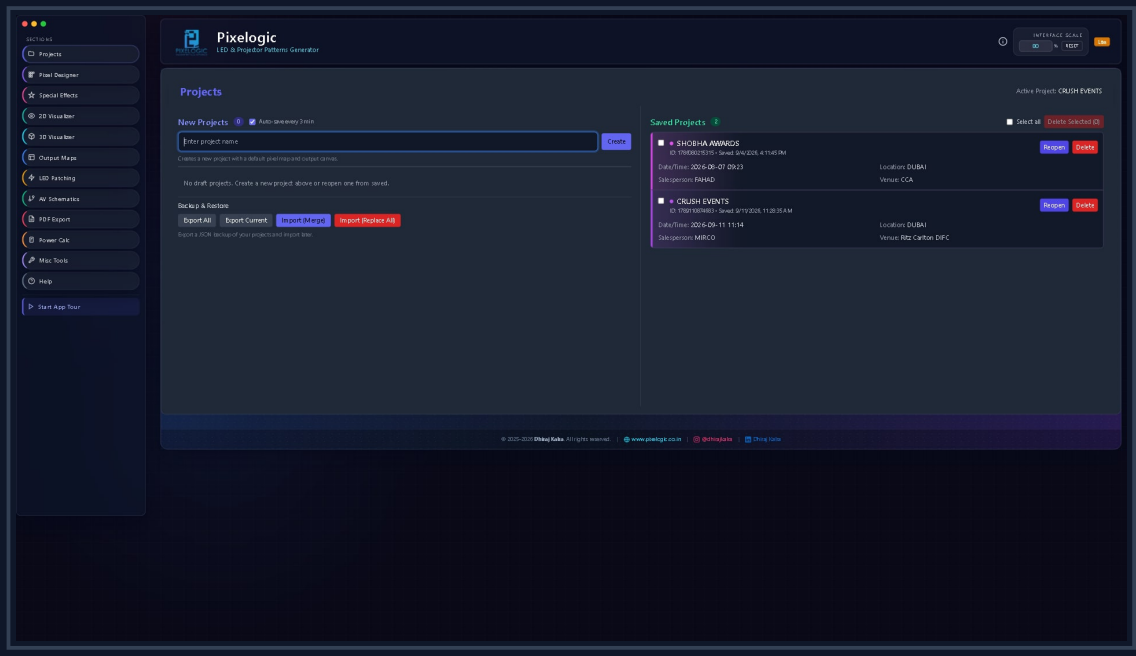
PURPOSE

Central hub for creating, organizing, and managing all your LED wall projects.

The Projects tab is your starting point in PixelLogic. Here you create new shows, manage existing drafts, and organize your saved projects. Each project contains all pixel maps, canvas layouts, effects, and export settings for a complete LED wall deployment.

Save project details in place without closing the editable project card. Auto-save is enabled by default every three minutes and can be disabled from the Projects tab. Saved Projects are for shows you explicitly archive, while JSON export/import provides portable backups.

The tab supports full project metadata including event name, date, location, venue, and salesperson. This information carries through to PDF exports and helps organize shows for multi-event operations.



1. Projects — Features & Workflow

Key Features

- Create new projects with custom names
- Edit project metadata (event date, location, venue, salesperson)
- Save project details in place without closing the project
- Optional auto-save every three minutes, enabled by default
- Duplicate existing projects
- Export individual projects as JSON backup files
- Import projects from JSON backup files
- Bulk export/import all projects
- Delete projects with confirmation
- Color-coded project cards for easy identification
- Auto-save to local storage

Typical Workflow

1. Click "New Project" and enter a project name
2. Fill in event metadata (date, location, venue)
3. Work on pixel maps, effects, and canvases in other tabs
4. Click "Save" to persist details while keeping the project open
5. Export as JSON for backup before delivery

2. Saved Projects

Finalized & Archived Shows

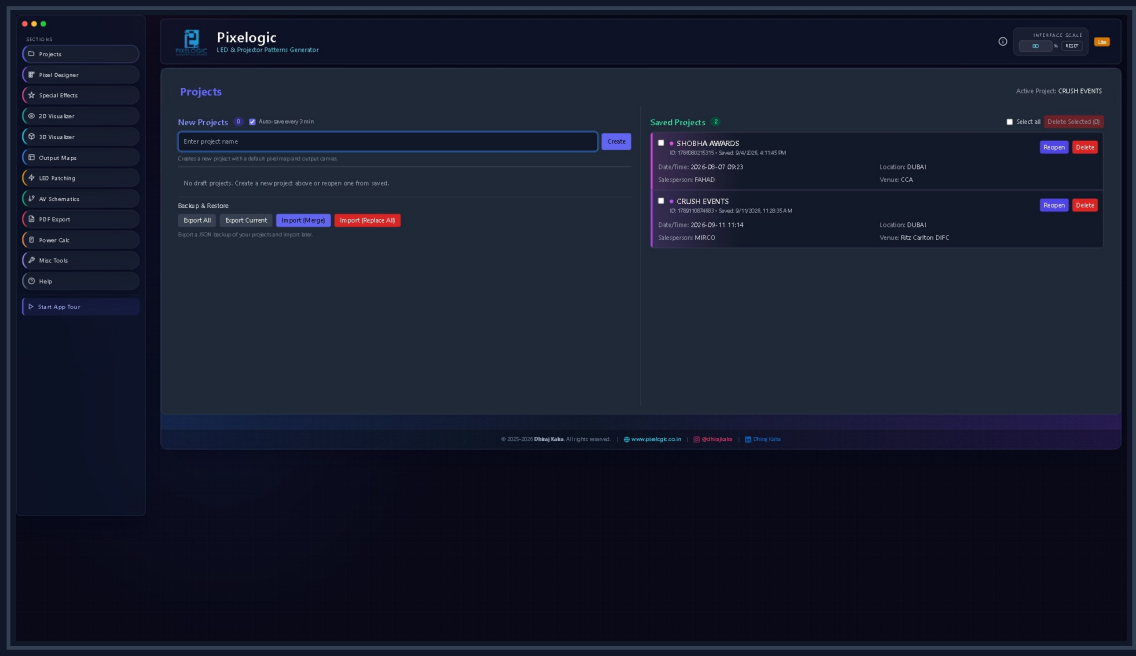
PURPOSE

View and manage all saved/finalized projects in a dedicated archive view.

The Saved Projects section displays all projects that have been marked as "saved" or finalized. These represent completed shows that are ready for deployment or have been archived after an event. The view provides a clean overview of your project library.

Each saved project displays its name, unique ID, save date, and event metadata. Projects can be reopened for further editing if modifications are needed, or permanently deleted when no longer required.

Multi-select functionality allows batch operations on saved projects, making it easy to manage large project libraries. Color coding helps visually distinguish between different projects at a glance.



2. Saved Projects — Features & Workflow

Key Features

- View all finalized/saved projects
- Multi-select with checkboxes for batch operations
- Reopen saved projects for further editing
- Delete saved projects permanently
- Color-coded project identification
- Display project metadata (name, ID, date, venue)
- Sort and filter projects

Typical Workflow

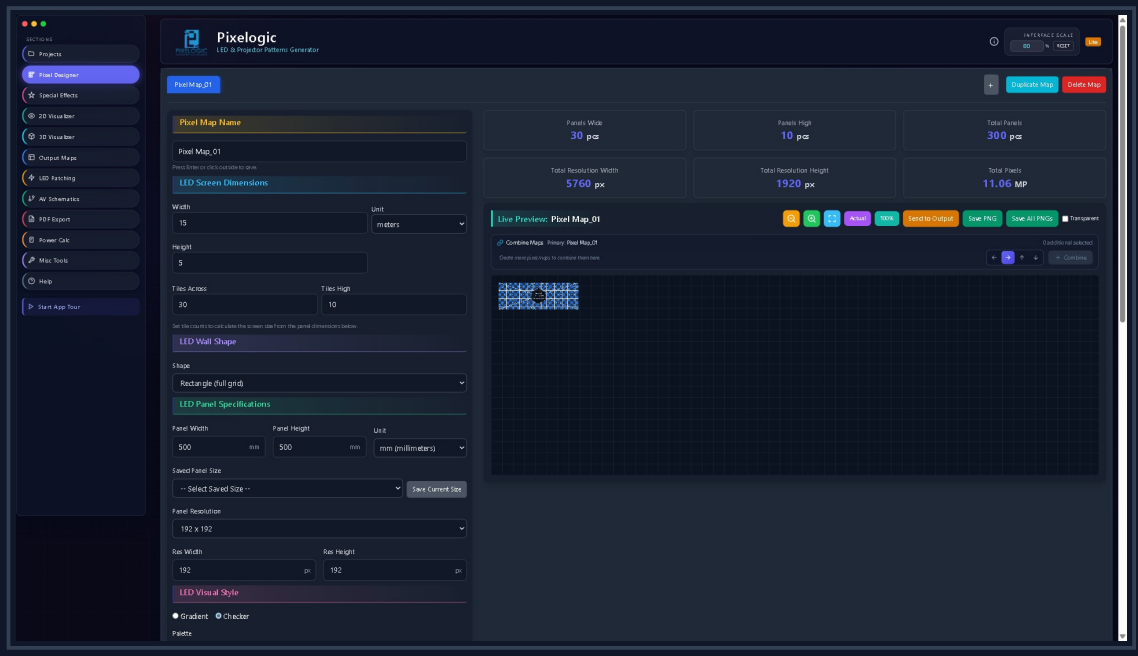
1. Navigate to the saved projects section
2. Review archived shows and their metadata
3. Select projects for batch operations if needed
4. Click "Reopen" to move a project back to drafts for editing
5. Use delete to permanently remove old projects

3. Pixel Designer

LED Wall Geometry & Configuration

PURPOSE

Design and configure LED pixel maps with precise panel dimensions, pitch, and resolution settings.



The Pixel Designer is the core design tool of PixelLogic. It allows you to define LED wall geometry by specifying panel dimensions (width × height in pixels), pixel pitch, and physical panel sizes. The designer calculates total resolution, physical dimensions, and generates visual representations of your LED walls.

Each project can contain multiple pixel maps representing different LED surfaces (main wall, side fills, floor, ceiling, etc.). The designer supports various visual styles including checker patterns, gradient fills, and solid colors to help visualize panel boundaries and pixel density.

Shape-based pixel maps let you build non-rectangular LED walls — Circle, Ellipse/Oval, Triangle, Diamond, Hexagon, Cross/Plus, or a fully custom hand-painted outline. Choosing a shape automatically disables the panels that fall outside it, yet the project still calculates as a rectangular grid so export, 3D, LED patching and power all keep working unchanged. For circular walls you can enter just a Radius, Diameter or Circumference and the wall size is computed for you — ideal when a client only gives one measurement.

Advanced features include panel gap simulation, automatic aspect ratios, rotation, and precise millimeter-level measurements. Screen dimensions and tile counts synchronize in both directions: Width updates Tiles Across, Height updates Tiles High, and changing tile counts recalculates the matching screen dimension. The calculated ratio is shown in reduced and decimal forms, for example 5:4 (1.25:1).

3. Pixel Designer — Features & Workflow

Key Features

- Define pixel map dimensions (width × height in panels)
- Two-way synchronization between screen dimensions and tile counts
- Set pixel pitch (mm) for accurate physical calculations
- Configure panel pixel resolution (e.g., 500×500, 250×250)
- Shape-based walls: Circle, Ellipse, Triangle, Diamond, Hexagon, Cross & Custom
- Size a circle by Radius, Diameter or Circumference — enter any one value
- Click-to-paint panels in the Live Preview for custom shapes & logos
- Disabled panels render transparent (no black background)
- Multiple visual styles: checker, gradient, solid, numbered
- Panel gap simulation for realistic visualization
- Multiple pixel maps per project
- Auto-calculate total resolution, physical size, pixel count
- Automatic reduced and decimal aspect ratio, such as 5:4 (1.25:1)
- Custom naming for each pixel map
- Duplicate and resize existing maps
- Real-time dimension preview
- Support for non-standard panel sizes
- Lock aspect ratio option

Typical Workflow

1. Select or create a pixel map from the map list
2. Enter panel count (columns × rows)
3. Pick a Wall Shape (or leave Rectangle) — for a circle, choose Radius / Diameter / Circumference and type one value
4. Set pixel pitch (e.g., 2.6mm, 3.9mm)
5. Define panel resolution (pixels per panel)
6. Choose visual style for the preview
7. Click panels in the Live Preview to fine-tune or paint a custom shape
8. Verify calculated dimensions match requirements
9. Add additional maps for multi-surface setups

4. Special Effects

Visual Overlays & Annotations

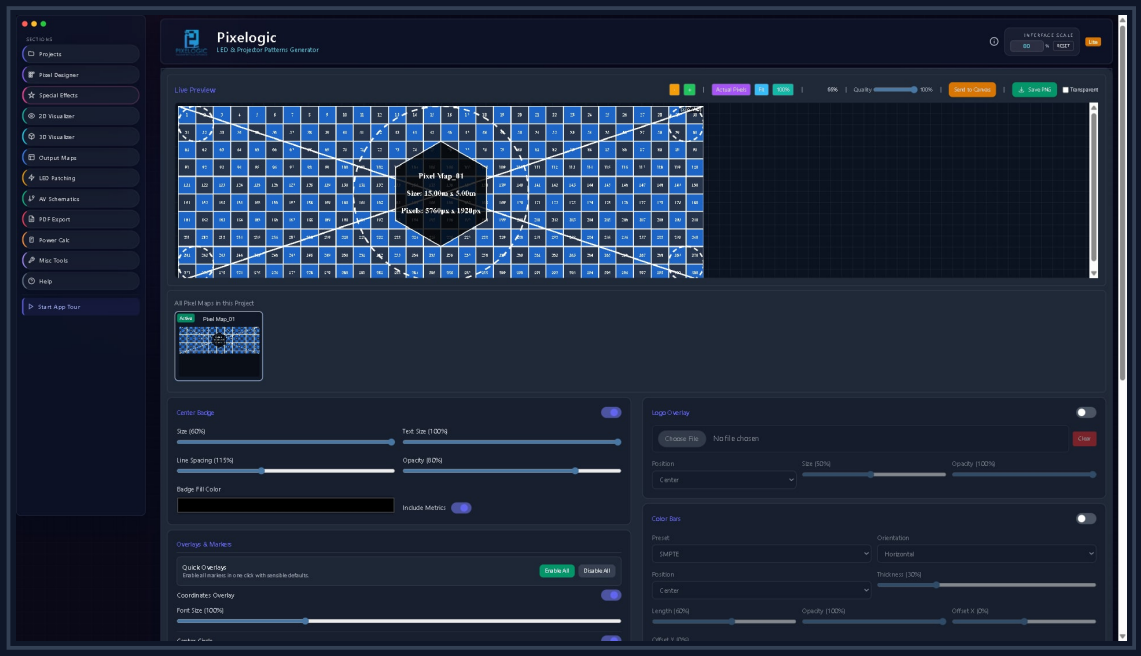
PURPOSE

Add visual overlay effects, text badges, markers, logos, and color adjustments to pixel maps.

The Special Effects tab provides tools for annotating and enhancing your pixel map visualizations. You can add text overlays showing resolution info, coordinate markers, dimension labels, and custom badges. Enable Include Metrics on the centre badge to add screen size, pixel resolution, and both aspect-ratio formats automatically. These effects help communicate technical specifications visually on exported images and PDFs.

Color adjustment tools allow you to modify the visual appearance of pixel maps with brightness, contrast, and tint controls. This is useful for creating presentation materials where different LED surfaces need distinct visual identification.

Logo placement, custom watermarks, and grid overlays round out the effects toolkit. All effects are non-destructive and can be toggled on/off independently, allowing you to create multiple export variations from the same base design.



4. Special Effects — Features & Workflow

Key Features

- Text overlay with resolution/dimension info
- Coordinate markers (X/Y axis labels)
- Custom text badges and labels
- Centre badge metrics include screen size, pixels and dual-format aspect ratio
- Logo/watermark placement
- Color bars and test patterns
- Brightness and contrast adjustment
- Grid overlay with customizable spacing
- Dimension lines and arrows
- Panel numbering overlay
- Non-destructive effect stacking
- Per-effect visibility toggle
- Effect presets for common configurations

Typical Workflow

1. Select the pixel map to add effects to
2. Choose effect type from the toolbar
3. Configure effect parameters (text, position, color)
4. Toggle effects on/off to preview
5. Export with effects applied to 2D Visualizer

5. 2D Visualizer

Full-Resolution Preview & Export

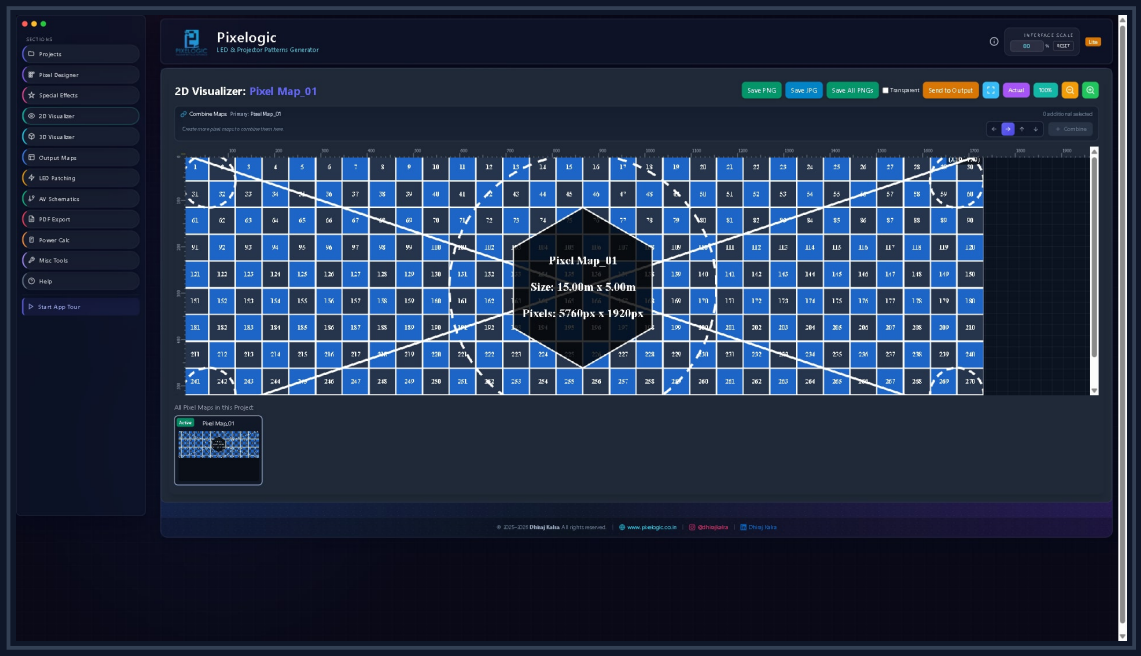
PURPOSE

Render full-resolution 2D previews of pixel maps and export as high-quality PNG images.

The 2D Visualizer renders your pixel maps at full resolution with all applied effects, providing an accurate preview of what the LED wall will look like. It supports zooming, panning, and measuring tools for detailed inspection of the design.

Export functionality allows saving the visualization as high-resolution PNG images suitable for client presentations, technical documentation, and print materials. You can export individual maps or combine multiple maps into a single composite view.

The visualizer respects all Special Effects settings, showing text overlays, dimension markers, and color adjustments as they will appear in the final output. This makes it the primary quality-check tool before generating PDFs or sharing designs with clients.



5. 2D Visualizer — Features & Workflow

Key Features

- Full-resolution pixel map rendering
- Zoom and pan controls
- Export as high-resolution PNG or JPG
- Batch export — Save All Pixel Maps as PNGs in one click
- Combine multiple maps in one view via a simple chip strip (click to add/remove, Right / Below / Left / Above / Grid placement with pixel gap)
- Per-map styles (gradient, palette, numbering, colour bars) preserved in composites
- Show/hide individual maps
- Background color customization
- Measurement and dimension tools
- Scale indicator
- Print-ready resolution output
- Effects preview integration

Typical Workflow

1. Navigate to 2D Visualizer tab
2. Select which pixel maps to display
3. Adjust zoom level for desired view
4. Toggle effects visibility as needed
5. Click Export to save as PNG image
6. Use exported image in presentations or documents

6. 3D Visualizer

Curved Arcs, 360° Cylinders & Stage Staging

PURPOSE

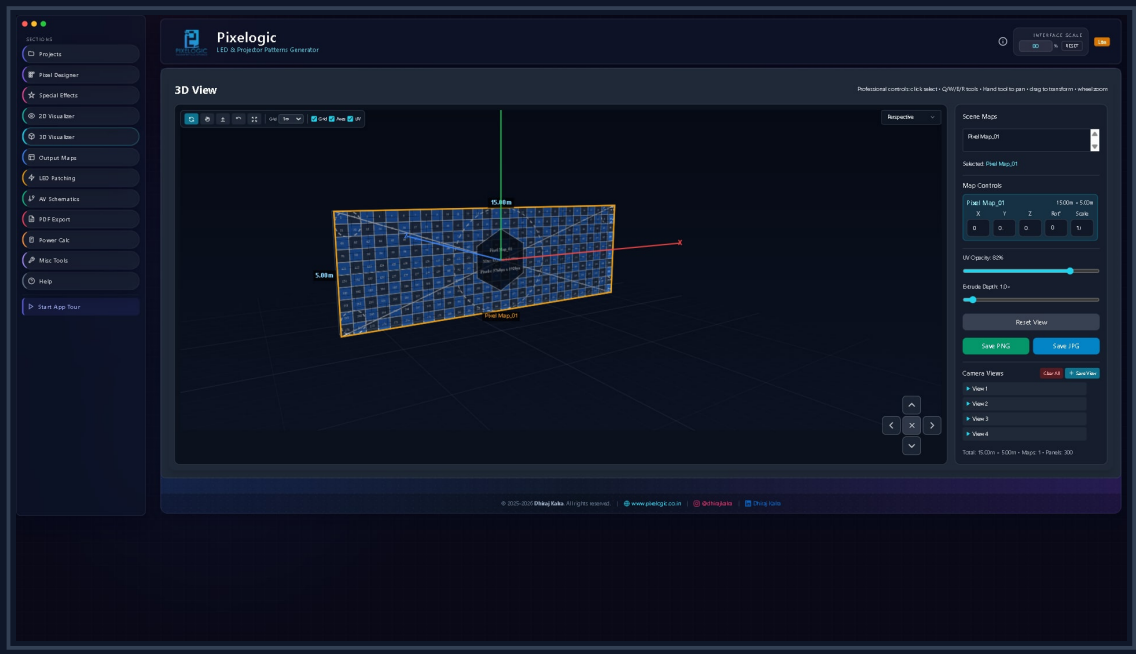
View flat, curved arc, and 360° cylindrical LED walls in an interactive 3D venue environment with Diameter & Radius sizing and magnetic snapping.

The 3D Visualizer provides an interactive perspective view of your LED wall designs. You can rotate, zoom, orbit, and pan around your LED surfaces, or switch each screen between Flat, Curved Arc (concave/convex), and full 360° Cylinder modes.

Curved Arc and 360° Cylinder modes display live Diameter (D) and Radius (R) readouts, arc angle (°), chord width, sagitta depth, and per-cabinet bevel angle (°/cab), along with 3D floor dimension lines for both Diameter and Radius. The Diameter & Radius Sizer computes the exact cabinet column count (Cols W / Rows H) from actual LED cabinet sizes.

Magnetic snapping aligns flat walls edge-to-edge and joins curved walls into a seamless continuous circular arc (Snap Arc Left / Right). The Stage Builder adds dimensioned stage decks beneath LED walls with Span to LED Bottom.

Every LED screen and stage deck has a high-contrast dimension callout. PNG and JPG exports retain these callouts while automatically omitting editor axes and transform gizmos.



6. 3D Visualizer — Features & Workflow

Key Features

- Flat, Curved Arc (concave/convex), and 360° Cylinder LED modes
- Live Diameter (D) and Radius (R) readouts and 3D floor dimension lines
- Cabinet-accurate Diameter & Radius Sizer with Cols (W) / Rows (H) steppers
- Magnetic edge-to-edge snapping and seamless Curved Arc continuation snapping
- Multiple camera angle presets and saved named camera views
- Per-map position / rotation / scale controls (W / E / R / Q gizmos)
- Direct stage dragging, Snap Below LED, and Span to LED Bottom actions
- Boxed screen and stage dimension callouts
- UV texture overlay with opacity and extrude-depth sliders
- Auto Revolve for hands-free presentations
- Clean PNG / JPG export without editor axes or transform gizmos
- Include 3D View Snapshot in the PDF Export

Typical Workflow

1. Open 3D Visualizer tab and select an LED screen in the Inspector
2. Choose Flat, Curved Arc, or 360° Cylinder mode
3. For Curved or Cylinder walls, enter Target Diameter (m) or Target Radius (m) and click Apply Columns
4. Use Snap to Target or Snap Arc Left / Right to join multi-wall arrays seamlessly
5. Add a stage deck and use Span to LED Bottom to align and width-match the stage
6. Save named camera views and export a clean PNG or JPG snapshot

7. Output Maps

Multi-Canvas Composition, Direct Add & Copy/Paste

PURPOSE

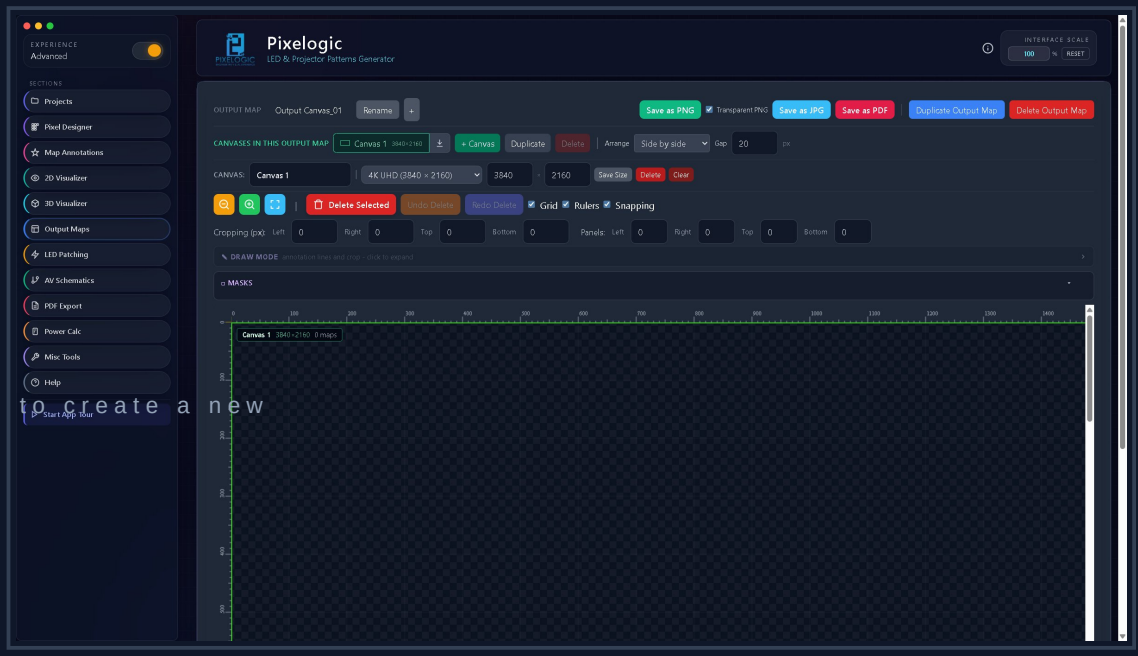
Arrange pixel maps into output canvases with direct map insertion, cross-canvas Copy & Paste, tile-snapped masks, and cropping.

The Output Maps tab is where you compose your final media server output configuration. Each Output Map can contain multiple named canvases representing physical video outputs, with a stable horizontal or vertical workspace layout.

You can add any project Pixel Map (or All Project Maps) directly from the Output Maps toolbar (+ Add to Canvas) without returning to Pixel Designer. You can also Copy (Ctrl+C) any selected map on a canvas and Paste (Ctrl+V) it into any other canvas, or click "Copy ! New Canvas" to create a new canvas and paste at (0, 0) in one click.

Selecting a canvas never changes its name or order. The active canvas remains in full color while inactive canvases stay visible in grayscale. Rename beside the top-level tabs edits the Output Map while nested canvas names remain independent.

Drawing tools enable adding reference lines, tile-snapped masks (Rect, Circle by Diameter & Radius, Ellipse, Polygon), and crops. Output maps remain linked to their source Pixel Designer map, and unchanged source images are cached for smooth drag/zoom redraws.



7. Output Maps — Features & Workflow

Key Features

- Direct + Add to Canvas Pixel Map picker right in the Output Maps toolbar
- Copy Map (Ctrl+C), Paste Map (Ctrl+V), Duplicate (Ctrl+D), an
- Multiple named canvases nested inside each Output Map
- Rename Output Maps independently of nested canvas names
- Cached source images and frame-coalesced drag/zoom redraws
- Stable canvas naming and order independent of selection
- Horizontal or vertical non-overlapping workspace layouts
- Active canvas in full color with inactive canvases in grayscale
- Direct mask selection with tile-snapped move/resize (Circle by Diameter & Radius); Shift precision for polygon vertices
- Source-linked map refresh that retains canvas placement and crop
- Canvas dimension presets (HD, 4K, custom) and panel-based cropping
- Save the selected canvas individually as PNG or export PNG/JPG/PDF/PPTX

Typical Workflow

1. Create or select an Output Map; use Rename, then Save or Enter
2. Add and name nested canvases independently in horizontal or vertical layout
3. Pick a Pixel Map from the toolbar dropdown and click + Add to Canvas
4. Use Copy Map (Ctrl+C) and Paste Map (Ctrl+V) or Copy !' duplicate across canvases
5. Position, crop by panels, and add tile-snapped masks as needed
6. Save an individual canvas as PNG or export the Output Map for documentation

8. LED Patching

Automatic & Manual Signal / Power Routing

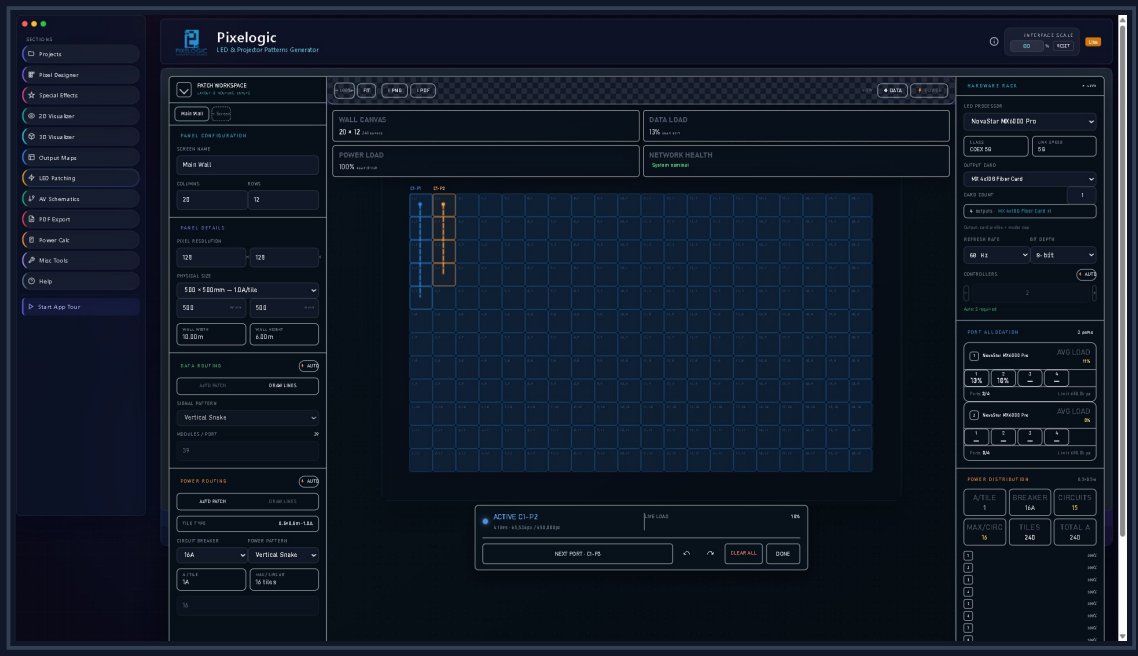
PURPOSE

Plan processor ports and power circuits automatically or draw every LED tile route by hand with live capacity feedback.

The LED Patching workspace combines display-array inputs, an interactive tile diagram, processor and output-card configuration, port allocation, and power distribution in one screen. Data and Power overlays can be viewed independently or together.

Auto Patch generates Vertical or Horizontal Snake/ZigZag routes using processor bandwidth, panel resolution, refresh rate, bit depth, tile amperage, and breaker capacity. The Hardware Rack shows controller requirements, used ports, per-port pixels, and load percentages.

Draw Lines mode lets you build the real installation route tile-by-tile. Click or drag through adjacent tiles, finish a line with Next Port or Next Circuit, and watch the enlarged dock report live pixels or amps and overload status. Undo, Redo, Clear All, and Done keep manual planning fast and recoverable.



8. LED Patching — Features & Workflow

Key Features

- Automatic Vertical/Horizontal Snake and ZigZag routing
- Manual click-or-drag routing across adjacent LED tiles
- Independent Data and Power routing modes and overlays
- Automatic C1-P1 onward port assignment with controller rollover
- Sequential power-circuit assignment
- Live active-line tile, pixel, amp, and load percentage feedback
- Overload and unassigned-path warnings
- Undo and Redo history for manual routes
- Clear All with undo recovery for the active route layer
- NovaStar/COEX processor and output-card profiles
- Refresh-rate and bit-depth-aware port capacity
- Automatic or fixed controller count
- Per-controller port allocation and load display
- Tile-aware breaker and circuit calculations
- FIT plus explicit zoom/pan controls without wheel-triggered resizing
- PNG and professional two-page PDF export

Typical Workflow

1. Set the wall columns, rows, panel resolution, physical size, tile type, and breaker rating
2. Select the processor, output-card profile, refresh rate, bit depth, and controller mode
3. For automatic routing, choose Auto Patch and select a Snake or ZigZag pattern
4. For manual signal routing, select Draw Lines under Data Routing and click or drag through adjacent tiles
5. Review live pixels and load percentage, then select Next Port to advance to the next C/P assignment
6. For manual power routing, select Draw Lines under Power Routing and use Next Circuit between feeds
7. Use Undo/Redo for corrections, Clear All to reset the active layer, and Done to leave drawing mode
8. Review port allocation, circuit load, and Network Health, then export PNG or PDF documentation

9. LED Master Sheets & CAD Drawings

Spreadsheet Scheduling & Selective 8 Architectural Vector Blueprints

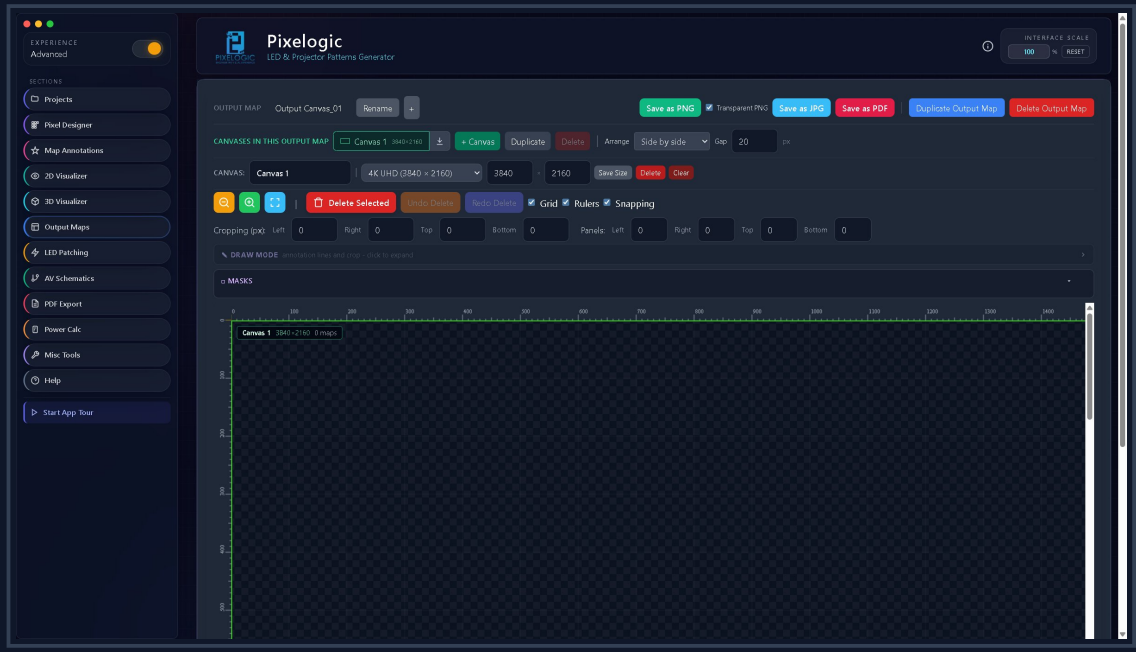
PURPOSE

Live spreadsheet tracking, multi-select batch deletion with scroll lock, and 8 selectable vector CAD engineering blueprints.

The LED Master Sheets module unifies high-level spreadsheet project scheduling with millimeter-accurate architectural CAD drawings. Pixel maps can only be deleted from this tab, supporting multi-row checkbox selection and batch Delete Selected (N) with horizontal/vertical scroll position preserved.

Three core views are provided: Project Schedule (live spreadsheet grid with streamlined on-screen Power in kW & Current in Amps, while full 3-phase/breaker/BTU details are reserved for Published PDFs), LED Catalog (30+ brands with verified pitch and electrical specs), and 8 CAD Drawings.

The 8 CAD blueprints include 1 Dimensions (mm/ft), 2 Raster Map, 3 Data Wiring, 4 Power Wiring, 5 System Flow, 6 Weight Estimate, 7 Heat Load (BTU), and 8 Bill of Materials. Use per-sheet checkboxes (All 8 / Core 6 presets or individual toggles) in both the sidebar and live Print/PDF preview window to choose which drawings appear in the final PDF.



9. LED Master Sheets & CAD Drawings — Features & Workflow

Key Features

- Live-synced Excel spreadsheet grid for all screens in project
- Multi-select checkboxes & batch Delete Selected (N) with scroll lock
- Streamlined on-screen Power (kW) & Current (A); full electrical specs in PDF
- 30+ brand LED technical database with custom brand/model creation
- 8 Vector CAD Sheets: Dimensions, Raster, Data, Power, System, Weight, BTU, BoM
- Selective PDF Sheet Checkboxes (All 8 / Core 6 presets + individual toggles)
- Live Print/PDF preview sheet filter bar to toggle drawings on the fly
- Publish All Screens Full Project PDF master package in one click
- Dedicated COMPANY LOGO column in CAD Title Block with persistent storage
- Dynamic page size selector (A1, A2, A3, A4, A5, Letter, Tabloid)
- Rigging hoist pickup points (WLL) & BTU/hr thermal dissipation calculations

Typical Workflow

1. Open the Sheets tab and review or edit project screens in the Schedule grid
2. Use row checkboxes and Delete Selected (N) to batch-remove maps without scroll jumps
3. Switch to the 8 CAD Drawings view and check/uncheck which sheets to include in the PDF (e.g. All 8 or Core 6)
4. Upload your company logo into the CAD Title Block and select paper size (A1–A5, Letter, Tabloid)
5. Click Print / Save PDF or Publish All Screens PDF and toggle sheet checkboxes live in the preview bar

10. PDF Export

Professional Report Generation

PURPOSE

Generate multi-page PDF reports with customizable templates, title blocks, and layout regions.

The PDF Export tab creates professional documentation packages from your PixelLogic designs. It generates multi-page PDF reports that include pixel map visualizations, technical specifications, dimension drawings, and project metadata in a polished presentation format.

Output canvases are emitted in stable order, with two canvases placed on each PDF or PowerPoint page when available. A final unpaired canvas receives its own page.

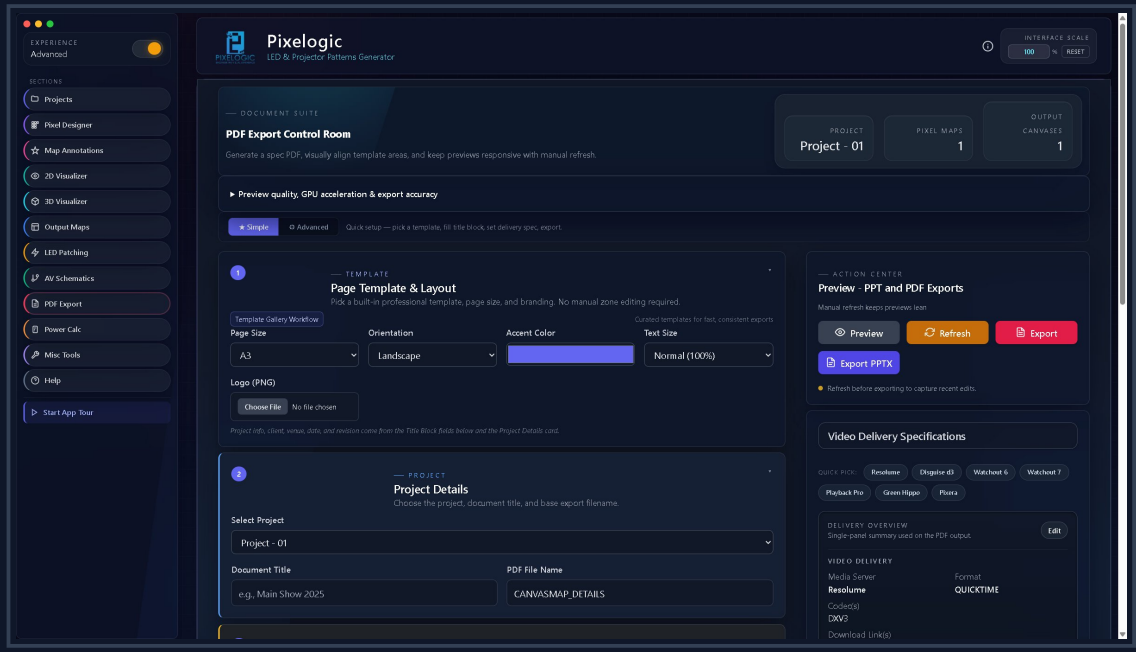
Advanced PDF export uses high-quality JPEG encoding for raster artwork and caps image resolution at 300 DPI based on its physical page placement. This substantially reduces file size while vector titles, detail text, and title blocks remain crisp.

Full/Lite preview quality does not reduce export source resolution. Pixel-map pages, output canvases and masks use full-resolution sources, native-pixel crops and full-quality annotation shadows. PDF encoding still follows the print-aware page limit.

The template system allows customizing report layouts with configurable regions, title blocks, headers/footers, and typography settings. You can define which pixel maps appear on which pages, add custom artwork backgrounds, and configure the information hierarchy.

Export options include page size (A4, A3, Letter, custom), orientation (landscape/portrait), quality settings, and selective page inclusion. Reports can include delivery specifications, cable schedules, power requirements, and any other project data — making this a one-stop documentation solution for LED installations.

Delivery presets for Resolume, Disguise d3, Watchout 6/7, Playback Pro, Green Hippo and Pixera synchronize Quick Pick, server and encoding settings. Simple and Advanced PDFs include clickable codec links; Resolume is labeled DXV3 CODEC.



10. PDF Export — Features & Workflow

Key Features

- Multi-page PDF report generation
- On-screen Preview quality, GPU acceleration & export accuracy guide
- Preview-independent source resolution, crops and annotation shadows
- Two output canvases per PDF or PPTX page when available
- Stable canvas order across report pages
- Print-aware 300-DPI raster cap with high-quality JPEG encoding
- Vector titles, details, and title blocks preserved for clarity
- Customizable page templates
- Title block with project metadata
- Header and footer configuration
- Selective pixel map inclusion
- Custom artwork/background support
- Page size options (A4, A3, Letter, custom)
- Landscape and portrait orientation
- Typography and font settings
- Technical specification tables
- Delivery specification pages with cleaner line spacing on the final page
- Advanced mode media server presets include Green Hippo and Pixera
- Quick-pick presets synchronize the Media Server dropdown and delivery details
- Pixel-map details include total pixel size and reduced + decimal aspect ratio
- Table of contents generation
- Company branding/logo placement
- Quality/resolution settings
- Batch export multiple projects

Typical Workflow

1. Read the preview/export accuracy guide; configure template and title block
2. Select which pixel maps to include
3. Confirm the nested output canvases and their stable order
4. Arrange page layout and regions
5. Add project metadata and specifications
6. Verify each map details block shows pixel size and aspect ratio
7. Choose a Video Delivery quick pick or Media Server dropdown; its server-specific details populate automatically
8. Preview the PDF layout; use Help > Check GPU for desktop diagnostics
9. Choose export settings (size, orientation, quality)
10. Generate and save the compact print-quality PDF or paired-page PPTX document

11. Power Calculator

LED Wall Power Consumption Analysis

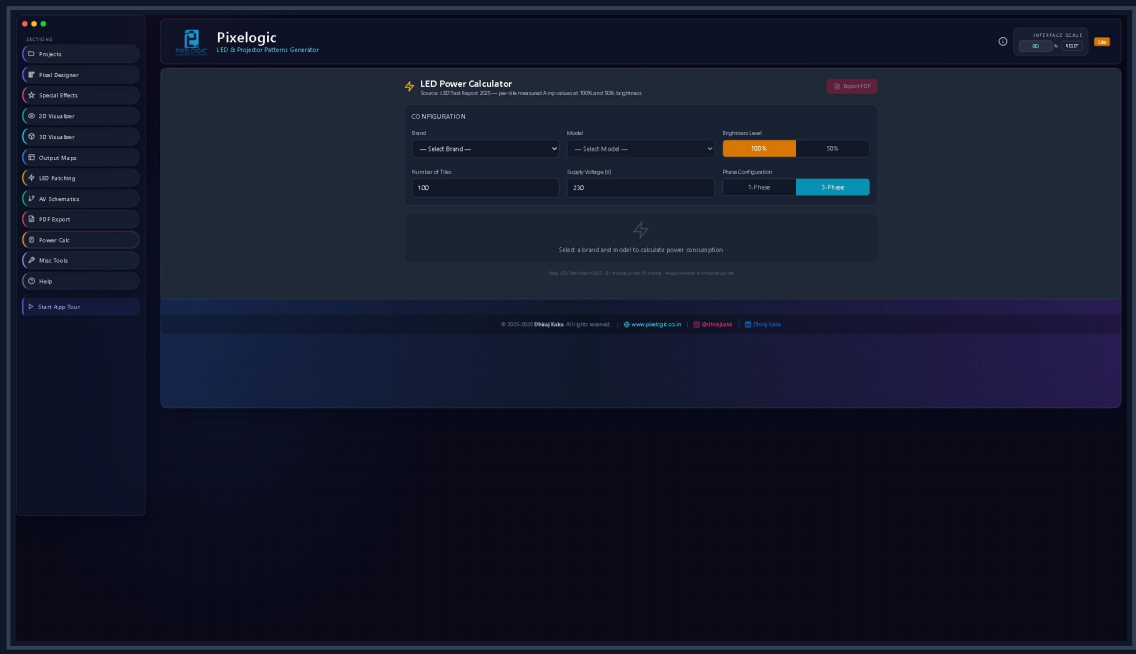
PURPOSE

Calculate LED wall power requirements using real bench-measured amperage data from 65+ panel models.

The Power Calculator uses real bench-measured amperage data from professional testing labs to accurately calculate power consumption for LED wall installations. The database contains 65+ LED tile models across 10 major brands (Gloshine, Absen, Unilumin, Infiled, and more).

For each selected model, the calculator provides per-tile amperage data across 5 color scenarios (Full Red, Full Green, Full Blue, Full White, and Standby) at both 100% and 50% brightness levels. It then scales these values based on your total tile count and voltage configuration.

The tool calculates total current draw, active power (kW), apparent power (kVA with power factor), and recommends breaker sizes with a 1.25× safety factor. For 3-phase configurations, it shows per-phase load distribution ensuring balanced power delivery. Results can be exported as a professional PDF power report.



11. Power Calculator — Features & Workflow

Key Features

- 65+ LED models across 10 brands
- Real bench-measured amperage data
- 5 color scenarios (R, G, B, White, Standby)
- 100% and 50% brightness levels
- Configurable tile count (1-9999)
- Supply voltage selection (100-480V)
- 1-Phase and 3-Phase configuration
- Active power calculation (kW)
- Apparent power calculation (kVA)
- Breaker sizing with 1.25× safety factor
- Per-phase load distribution analysis
- IEC standard breaker size recommendations
- Professional PDF power report export
- Color-coded scenario comparison cards

Typical Workflow

1. Select LED brand from dropdown
2. Choose specific model (auto-populated based on brand)
3. Set brightness level (100% or 50%)
4. Enter total number of tiles
5. Configure supply voltage (default: 230V)
6. Select phase configuration (1-phase or 3-phase)
7. Review power summary and scenario breakdown
8. Export power report as PDF if needed

12. Utility Tools

Circle/Cylinder Calculator, Video Converter, Aspect Ratio & LTC

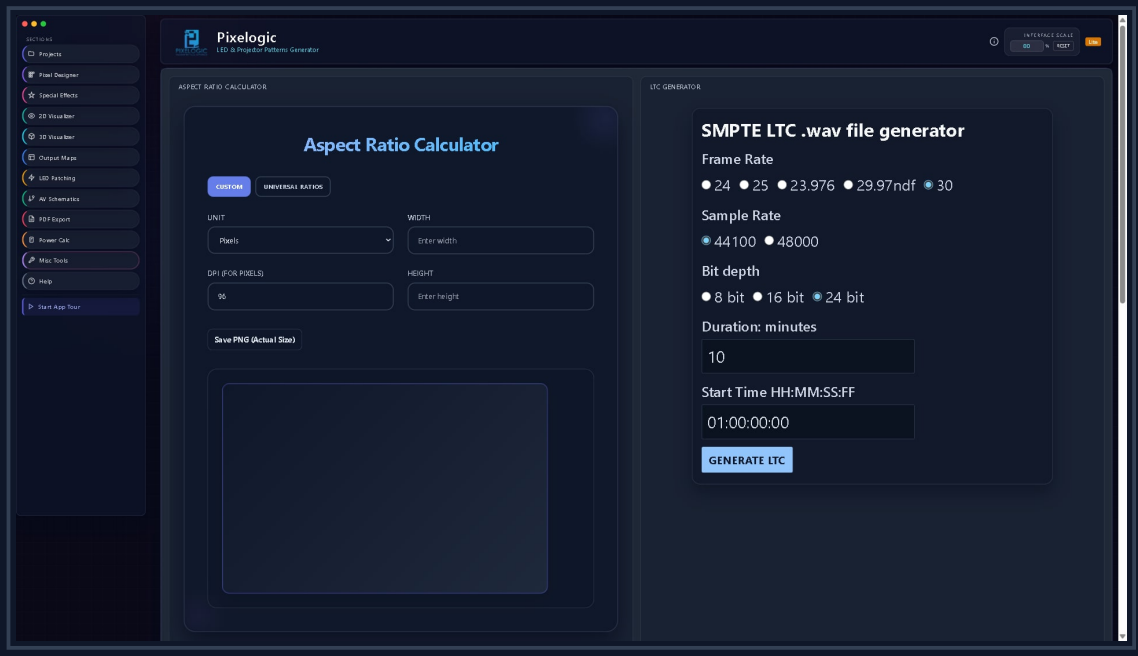
PURPOSE

Circle/Cylinder Diameter & Radius calculator, ProRes/DNxHD Video Converter, Aspect Ratio Calculator, and LTC Generator.

The Utility Tools tab provides four specialized engineering utilities for LED design and live production: a Circle / Semi-Circle / Cylinder LED Calculator, a ProRes/DNxHD/H.264/H.265 Video Converter, an Aspect Ratio Calculator, and a Linear Timecode (LTC) Generator.

The Circle / Semi-Circle / Cylinder Calculator computes exact cabinet counts, chord/arc lengths, angles, and both Diameter (D) and Radius (R) in meters and feet from cabinet width and pitch. The Video Converter transcodes media server files with live progress and target bitrate control.

The Aspect Ratio Calculator offers Custom resolution scaling and Universal Ratios reference cards. The LTC (SMPTE Timecode) Generator produces synchronized timecode audio signals across standard frame rates (23.976, 24, 25, 29.97, 30 fps) for media servers and show control.



12. Utility Tools — Features & Workflow

Key Features

- Circle, Semi-Circle & 360° Cylinder LED calculator
- Live Diameter (D) and Radius (R) readouts in meters & feet
- Cabinet count, arc length, and inter-cabinet angle calculation
- Built-in Video Converter (ProRes, DNxHD, HAP, H.264, H.265)
- Aspect Ratio Calculator — Custom mode & Universal Ratios reference
- Calculate ratio from any resolution and scale proportionally
- LTC/SMPTE Timecode Generator (23.976, 24, 25, 29.97, 30 fps)
- Real-time timecode display and audio output

Typical Workflow

1. Select the desired utility sub-tab (Aspect Ratio, Circle/Cylinder, Video Converter, or LTC)
2. For Circle/Cylinder: Enter cabinet width and count or target Diameter (D) / Radius (R)
3. For Video Converter: Select input media, choose codec preset, and convert
4. For Aspect Ratio: Enter resolution or select a standard ratio preset
5. For LTC: Configure frame rate and start time to generate SMPTE audio

13. Projector Test Pattern

Projection Calculator & Test Patterns

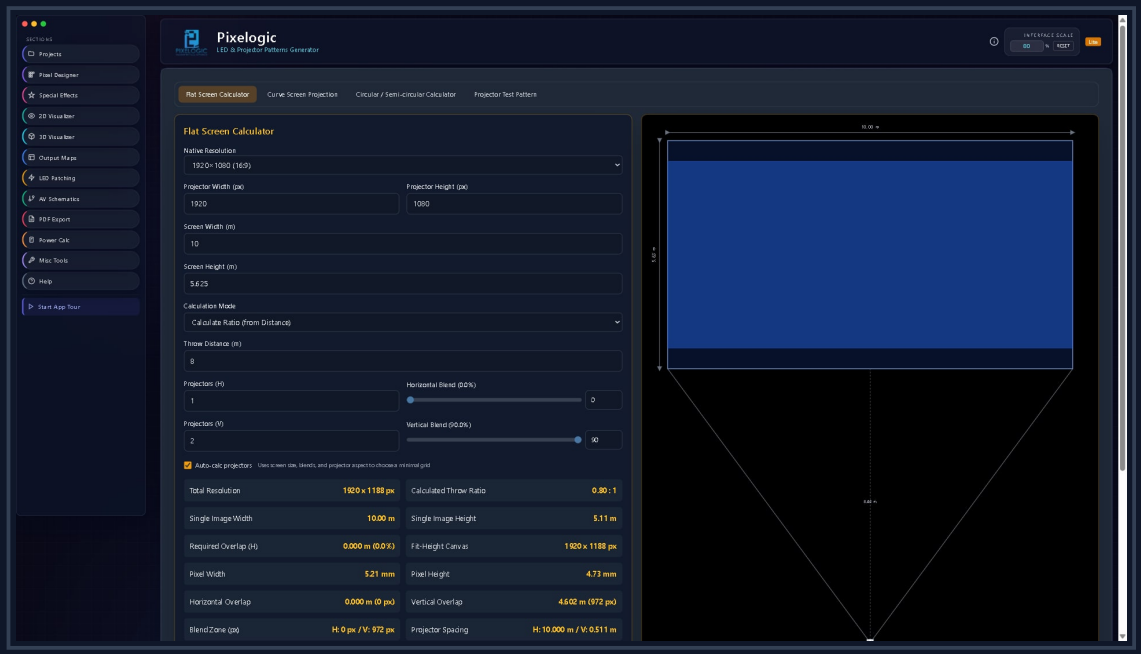
PURPOSE

Generate test patterns and calculate projection parameters for flat, curved, and circular surfaces.

The Projector Test Pattern tool is PixelLogic's projection mode centerpiece. It generates professional test patterns for projector alignment, edge-blending calibration, and geometry correction. The tool includes comprehensive calculators for three projection surface types: flat, curved, and circular/dome.

The Flat Calculator computes throw distances, image sizes, lens shift requirements, and multi-projector overlap zones for standard flat-screen installations. The Curved Calculator handles cylindrical and custom-curve surfaces with wrap angle calculations. The Circular/Dome Calculator addresses planetarium-style and immersive dome projections.

Test pattern generation includes grid patterns, crosshatch, color bars, focus charts, geometry test images, and custom resolution patterns. All patterns can be displayed fullscreen on connected displays for real-time projector calibration during installation.



13. Projector Test Pattern — Features & Workflow

Key Features

- Test pattern generation (grid, crosshatch, color bars, focus)
- Fullscreen pattern display on connected monitors
- Custom resolution pattern generation
- Flat projection calculator (throw distance, image size)
- Lens shift calculation
- Multi-projector overlap zone calculation
- Curved surface projection calculator
- Wrap angle and surface radius computation
- Circular/dome projection calculator
- Projector specification database
- Edge blend overlap visualization
- Geometric correction reference patterns

Typical Workflow

1. Select projection surface type (flat/curved/dome)
2. Enter projector specifications (lens, throw ratio)
3. Configure surface dimensions
4. Calculate throw distance and image size
5. For multi-projector: configure overlap and blending
6. Generate appropriate test pattern
7. Display fullscreen on projector for calibration

13. Projector Test Pattern — Subtabs

Tab 1

Settings

Base projector and surface inputs, pattern type selection, rendering options

Tab 2

Flat Calculator

Standard flat-surface projection: throw ratio, distance, multi-projector layouts

Tab 3

Curved Calculator

Cylindrical/curved surface projection calculations with wrap angle

Tab 4

Circular/Dome Calculator

Dome and circular projection geometry, coverage calculations

14. Rendering & GPU

Sharper Previews, Smoother Editing

PURPOSE

Choose preview quality and inspect actual desktop graphics acceleration.

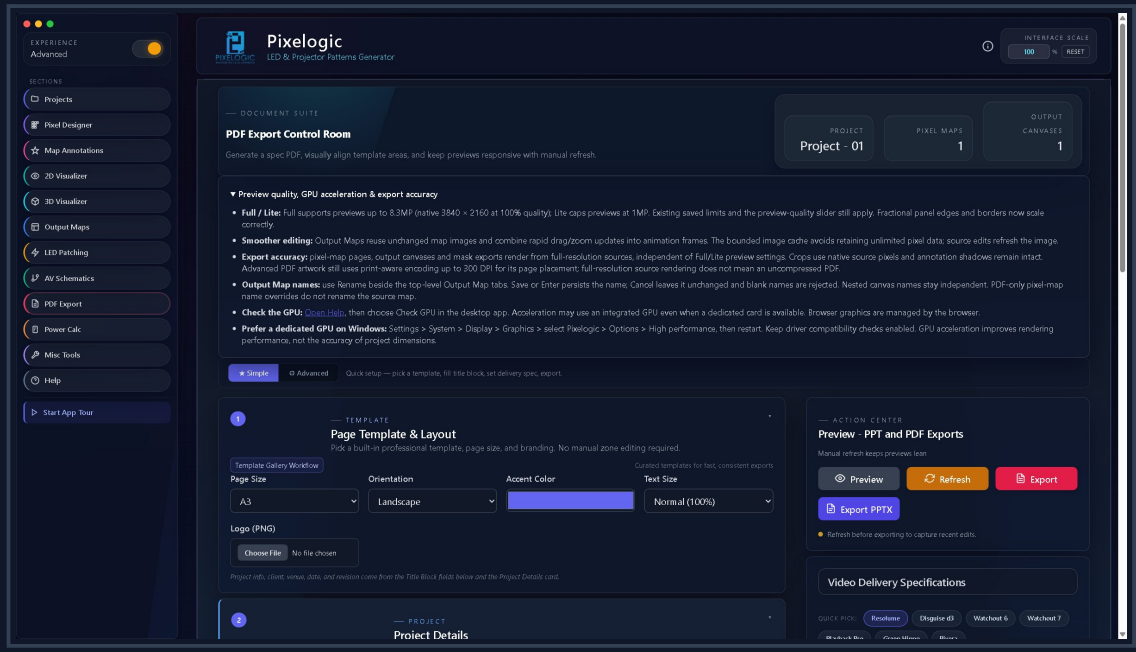
Full supports previews up to 8.3MP: native 3840 x 2160 at 100% quality. Lite caps at 1MP. Existing saved limits and the quality slider still apply; very large maps may be downscaled.

Panel boundaries share rounded edges at fractional scales and borders scale proportionally. Output Maps reuse unchanged source images and combine rapid drag/zoom requests into animation frames.

The bitmap cache retains at most 64MiB of pixel data and 128 entries. Oversized images render without being cached; source, logo and font changes refresh content. Coordinate overlays and composites render directly.

Desktop acceleration is enabled by default subject to driver compatibility checks. Help > Check GPU reports the active device, renderer, detected GPUs and Canvas/compositing/rasterization status. A dedicated GPU may be available but not active.

GPU acceleration improves performance, not the accuracy of project dimensions. Full-resolution export sources remain independent of preview mode; Advanced PDF artwork still follows its print-aware 300-DPI limit.



14. Rendering & GPU — Features & Workflow

Key Features

- Full: up to 8.3MP; Lite: 1MP pixel-map previews
- Native 4K at 100% preview quality in Full mode
- Consistent panel edges and border widths at fractional scales
- Bounded source-image cache for Output Maps
- Drag and zoom redraws combined into animation frames
- Source edits, logo loading and font changes refresh images
- Native-pixel crops and full-resolution export sources
- Full-resolution annotation shadows independent of Lite mode
- Check GPU: actual desktop device and acceleration status
- Browser mode explicitly identifies browser-managed graphics
- Driver compatibility checks remain enabled

Typical Workflow

1. Choose Full in Help for sharper previews, or Lite for lower preview load
2. Set preview quality to 100% when inspecting a native 4K map
3. Open Help > Check GPU in the desktop app
4. Compare the active GPU/renderer with detected devices
5. On Windows: Settings > System > Display > Graphics
6. Select Pixelogic > Options > High performance; restart the app
7. Check GPU again; the OS and driver determine the active device
8. For driver troubleshooting only: launch with PIXELLOGIC_DISABLE_GPU=1. Remove the override and restart to restore normal selection

15. External Output

Edit and Present Multiple Maps

PURPOSE

Arrange multiple maps directly on the external monitor, then show clean output.

Desktop only: connect another monitor, select Win+P > Extend, then open External Output in the header. The editor monitor is excluded; choose a connected secondary display.

Select a pixel map or nested output canvas. Save PNG in Edit Mode exports the full canvas with maps aligned together to the top-left and exported X/Y coordinates. Spacing and live placement stay unchanged.

Fit preserves aspect ratio with black letterboxing. 1:1 maps one source pixel to one physical display pixel, accounting for Windows scaling. Content is centered; oversized sources are clipped.

Edit Mode supports Add map, dragging, corner handles, whole-pixel W/H and Scale X/Y percentages (100% = native resolution). Snap aligns edges/centers; Alt bypasses it. The top-right badge shows the working output size. Finish editing hides controls.

Layout edits persist and affect canvas exports, without changing LED panel specifications. Live source updates run up to 10 times per second. Freeze and Blackout disable editing; display options do not change export settings. This is not genlocked playback.



15. External Output — Features & Workflow

Key Features

- One external display and one selected source per session
- Pixel-map sources become saved editable output canvases
- Full-resolution PNG with exported X/Y coordinates, independent of Full/Lite
- New maps start at 100%; 1:1 is default; Fit never changes saved scale
- Live / Freeze, independent Blackout, Stop and output Escape
- Optional annotations/logos and canvas drawings
- Add multiple independent instances of project pixel maps
- Drag positioning, corner resizing and precise X/Y/W/H values
- Keep aspect or free resizing; saved scale survives source refreshes
- Snap with alignment guides, output-resolution badge and pixel/percentage sizing
- Bottom toolbar: Hide controls / Show controls; Finish editing hides all overlays
- Masks: Ignore, Show fills, Inside only or Outside only
- Overlapping clipping masks form a union; no masks means no clipping
- Stops on disconnect, display reconfiguration, project change or source deletion
- No automatic restart or fallback onto the editor monitor
- Limits: 16384px per side, 64MP per surface, 128MP source budget, 500,000 panels per map

Typical Workflow

1. Connect a display and choose Win+P > Extend
2. Open External Output; Refresh displays if necessary
3. Select the external display and pixel map or nested canvas
4. Choose Fit or 1:1 and explicit annotation, drawing and mask options
5. Start output, then Close controls to continue editing live
6. Enable Edit Mode to add, select, drag or resize maps on the monitor
7. Use Snap for edge/center alignment; hold Alt to bypass snapping
8. Enter W/H in whole pixels or Scale X/Y in percent; 100% is native resolution
9. Keep aspect uses native proportions; 100% restores exact size; Shift resizes freely
10. Choose Finish editing to hide handles and the toolbar
11. Reopen controls to Freeze, Resume live or toggle Blackout
12. Stop before changing the source or display; Escape in output also stops
13. After unplugging or changing resolution, scaling or arrangement, select the display and restart explicitly
14. Save PNG: top-left alignment, full canvas size, unchanged live layout; oversize clips

16. Help & Reference

Documentation, Interactive App Tour & System Scaling

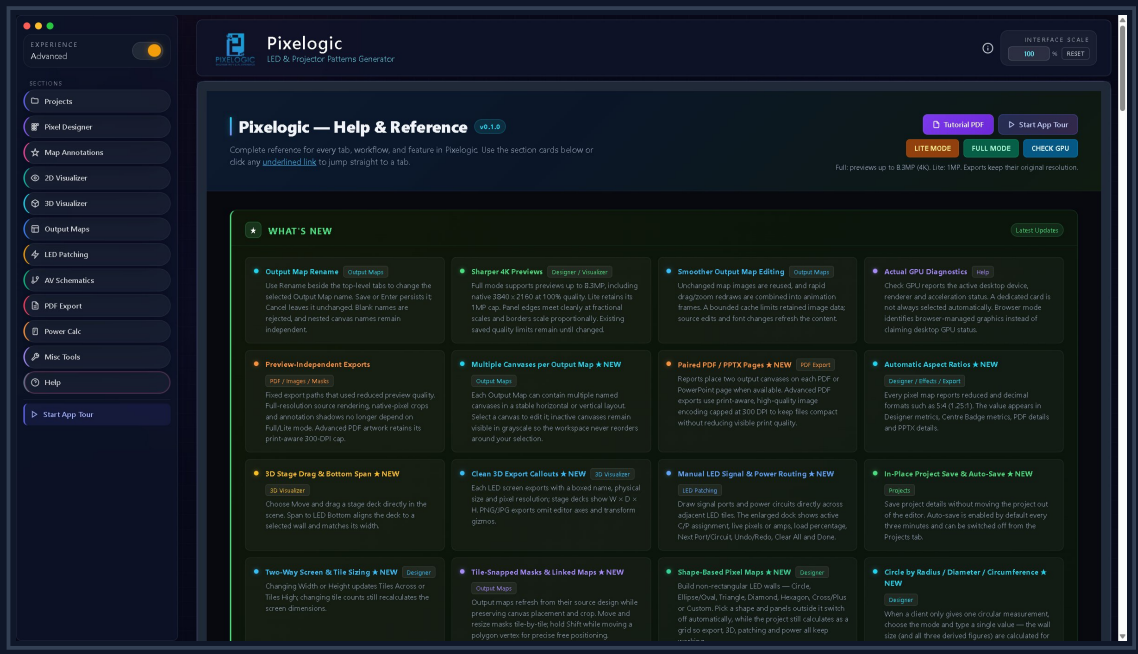
PURPOSE

In-app documentation, interactive Start App Tour, Tutorial PDF launcher, GPU diagnostics, and Interface Scale (Auto-Fit).

The Help tab provides comprehensive in-application documentation covering all features, keyboard shortcuts, workflow guides, and troubleshooting information. You can launch the interactive Start App Tour or open this full Tutorial PDF directly from the Help header.

The "What's New" section highlights the latest updates: Selective CAD PDF Sheet Checkboxes (All 8 / Core 6 + live preview filter), Direct Output Maps Add/Copy/Paste & Copy to New Canvas, 3D Cylinder & Curved Arc Diameter (D) + Radius (R) with Cols W / Rows H sizer, and Multi-Select Map Deletion with scroll lock.

Header INTERFACE SCALE (AUTO-FIT) automatically scales the workspace to fit your display resolution without unwanted scrollbars, or lets you pick a fixed zoom (75%–110%). Full mode supports previews up to 8.3MP and Lite caps at 1MP; Check GPU inspects active desktop hardware acceleration.



16. Help & Reference — Features & Workflow

Key Features

- Complete feature documentation & What's New changelog
- Interactive Start App Tour walkthrough covering all latest workflows
- One-click Tutorial PDF (Full Guide) launcher from Help header
- Header INTERFACE SCALE (AUTO-FIT) for any display resolution
- Master keyboard shortcuts reference (including Output Maps Ctrl+C / Ctrl+V / Ctrl+D)
- Full (8.3MP) / Lite (1MP) pixel-map preview modes
- Check GPU with active renderer and detected device details
- Troubleshooting guides & Windows High Performance GPU setup
- Export format specifications & offline operation reference

Typical Workflow

1. Open Help tab or click the header ? icon for quick reference
2. Click Start App Tour for an interactive guided walkthrough of all modules
3. Click Tutorial PDF (Full Guide) to open the complete illustrated manual
4. Use Header INTERFACE SCALE (AUTO-FIT) to fit the UI cleanly to your monitor
5. Choose Full for 4K detail or Lite for lower preview memory load
6. Run Check GPU to verify hardware acceleration on your workstation

Quick Reference — All Tabs at a Glance

1. **Projects**
Central hub for creating, organizing, and managing all ...

2. **Saved Projects**
View and manage all saved/finalized projects in a dedic...

3. **Pixel Designer**
Design and configure LED pixel maps with precise panel ...

4. **Special Effects**
Add visual overlay effects, text badges, markers, logos...

5. **2D Visualizer**
Render full-resolution 2D previews of pixel maps and ex...

6. **3D Visualizer**
View flat, curved arc, and 360° cylindrical LED walls i...

7. **Output Maps**
Arrange pixel maps into output canvases with direct map...

8. **LED Patching**
Plan processor ports and power circuits automatically o...

9. **LED Master Sheets & CAD Drawings**
Live spreadsheet tracking, multi-select batch deletion ...

10. **PDF Export**

13. **Projector Test Pattern**
Generate test patterns and calculate projection paramet...

14. **Rendering & GPU**
Choose preview quality and inspect actual desktop graph...

15. **External Output**
Arrange multiple maps directly on the external monitor,...

16. **Help & Reference**
In-app documentation, interactive Start App Tour, Tutor...

Pro Tips

- Use Ctrl+S to quick-save your project at any time
- Switch between Pixelogic and Projection Logic modes using the mode toggle in the header
- Export your work frequently — use JSON backup for complete project preservation
- The Power Calculator uses real lab-measured data, not manufacturer specifications
- All tools work offline — no internet connection required after installation